# JDCNet Uses Confidence Gated Privileged Modality Distillation for Cost Preserving X Ray Inference

Bo Ma[a,*], Wei Qi Yan[a], Jinsong Wu[b], Kun Liu[c]

[a] *Auckland University of Technology, Auckland, 1024, New Zealand*
[b] *Guilin University of Electronic Technology, Guilin, China*
[c] *Hebei University of Technology, School of Artificial Intelligence and Data Science Tianjin, China*

**Abstract**

Training-only privileged-modality learning uses an expensive modality during training while keeping a single-modality deployment path. We study chest radiograph (X-ray) classification with same-patient computed tomography (CT) available only during training. JDCNet adds an auxiliary CT-teacher target, the teacher's argmax label or its softmax distribution through a Kullback–Leibler term (soft-KL), only on samples whose teacher confidence exceeds a threshold; the deployed model has the same ResNet-18 computation graph as the supervised baseline. On 510 same-patient pairs from the public BIMCV-COVID19 releases of the Valencian Region Medical ImageBank (113 COVID-19, 397 non-COVID), with patient-level five-fold cross-validation and three seeds, 15 of 16 adaptively explored JDCNet configurations improved mean balanced accuracy (BA) over the supervised baseline (average +0.020), but none remained significant after Holm adjustment. In a pre-registered experiment with 240 batch-matched supervised restarts on the same splits, batch size had no detectable effect on the baseline (BA 0.620 versus 0.625), and when both methods selected one of 16 runs per cell on the validation partition, JDCNet exceeded supervised training by +0.037 BA (95% confidence interval +0.016 to +0.058; patient-level +0.007 to +0.068). Logit distillation and contrastive alignment did not help. Teacher diagnostics indicate that the gate acts mainly as CT-informed sample weighting. All models, including JDCNet, performed at chance on an external radio-

*Corresponding author.
*Email address:* amabo1215@gmail.com (Bo Ma)

graph set. Confidence-gated CT supervision thus gives a small within-cohort advantage on one public cohort and a binary task; generalization requires independent paired replication.

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- **Corresponding author:** Bo Ma (amabo1215@gmail.com), Auckland University of Technology, Auckland 1024, New Zealand.